

EART11200 - The Natural Scientist's Toolkit
Mathematics and Physics

Week 10 Lecture - Vectors and Partial Differentiation

LAST WEEK...

- **Second-order ordinary differential equations** are ODEs that involve a **second-order derivative**.
- We could devote an entire course looking at solving different types of second-order ODEs, but we will only concentrate on

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = 0$$

where a, b and c are constants.

- This is referred to as a **homogeneous, linear, second-order ODE**.
- Note that we can write the exact same expression using alternative notation as per the following equation:

$$ay'' + by' + cy = 0$$

Last Week...

- To solve these second-order ODEs we first write out the **auxiliary** equation:

$$a\lambda^2 + b\lambda + c = 0$$

- Find roots of the quadratic equations using:

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- If λ_1 and λ_2 are real and not equal ($\lambda_1 \neq \lambda_2$ then we can write a general solution as:

$$y = Ae^{\lambda_1 x} + Be^{\lambda_2 x}$$

- If our two roots are real and equal (i.e. $\lambda_1 = \lambda_2$) then we use the formula:

$$y = (A + Bx)e^{\lambda_1 x}$$

Last Week...

- To find the values of the constants A and B, we use information such as $y = 0$ and $y = 1$ when $x = 0$.
- These are called **initial conditions**, and they tell us the value of the solution and/or its derivatives at a specific point, allowing for a unique solution to the differential equation.
- This will lead us to solving simultaneous equations.
- Once we

Last Week...

- Fick's and Fourier's Laws describe the transfer of mass and heat.
- If we assume steady state, we get second-order ODEs which we can simply integrate twice

$$\frac{d^2T}{dx^2} = 0 \implies T(x) = Ax + B$$
$$\frac{d^2C}{dx^2} = 0 \implies C(x) = Cx + D$$

- We can use boundary conditions to solve this and find an expression.
- **Boundary value problems** differ from **initial value problems** as we specify upper and lower boundary conditions are specified (i.e, when $x = x_0$ and $x = x_1$).
- This gives us more information this means that we have to make the solution meet at both points.

Course layout

Date	Week	Topic
27/01/2025	20	Calculating Gradients
03/02/2025	21	Introduction to Differentiation
10/02/2025	22	Differentiating Combinations of Functions
17/02/2025	23	Optimisation & Properties of Derivatives
24/02/2025	24	Second Order Derivatives & Root Finding
03/03/2025	25	Introduction to Integration
10/03/2025	26	Integration using Newtons Laws
17/03/2025	27	Integration by Substitution
24/03/2025	28	Integrating Second Order ODEs
31/03/2025	29	Vectors and Partial Differentiation
07/04/2025	30	Non-teaching week
14/04/2025	31	Easter break
21/04/2025	32	Easter break
28/04/2025	33	Revision: Differentiation
05/05/2025	34	Revision: Integration
Exam Period		

Intended Learning Outcomes

(1) Describe the properties of scalar and vector fields and their applications in mathematics and physics

(2) Apply basic vector operations, such as addition, subtraction, scalar multiplication, and magnitude, in theoretical and real-world applications.

(3) Compute partial derivatives to determine rates of change of scalar fields in theoretical and real-world applications.

Working in More Than One Dimension

- The topics Differentiation (Weeks 1-5) and Integration (Weeks 6-9) were centred around a function in one variable or dimension

$$y = f(x)$$

- We differentiated or integrated in terms of that one variable:

$$\frac{dy}{dx} \text{ or } \int f(x)\, dx$$

- In many cases, we may need to work in more than one dimension (e.g. in space)

$$f(x, y) = x^2 + y^2$$

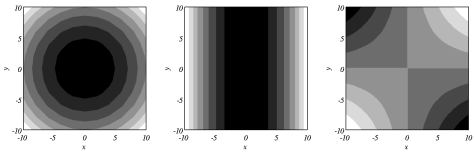
- So, we conclude the course by looking at **scalar fields**, **vectors** and **partial differentiation**.

Scalar Fields

- A **scalar field** is a function of two or more variables

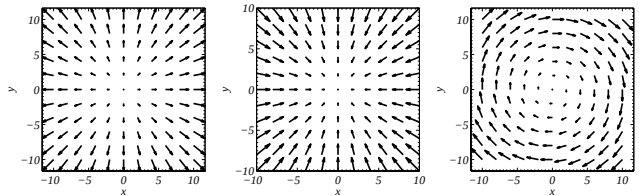
$$f(x, y) = x^2 + y^2$$

- Scalar fields represent a single value at different positions.
- They have **magnitude but no direction** and can be represented with a colour plot.
- For example, wind speed – but not direction.
- In essence, they are functions in higher dimensions than we have seen thus far.



Vector Fields

- A **vector field** is a mathematical function that assigns a vector to each point in space.
- For example, wind speed – but not direction.
- They have **magnitude and direction** as we can determine a direction, or flow and the magnitude from the length of the arrow.

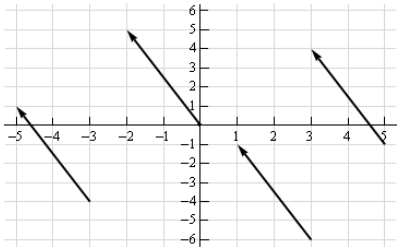


Vector Notation

- Each line in this figure represents the same vector, but a different starting position.
- Wherever the starting position is, each vector moves 2 units to the left and 5 units up.
- If we write this procedure in vector form we write:

$$\vec{v} = \langle -2, 5 \rangle$$

- Notice the distinction in style between the variable and the brackets.
- Specifically, $\langle -2, 5 \rangle$ is fundamentally different from writing $(-2, 5)$.

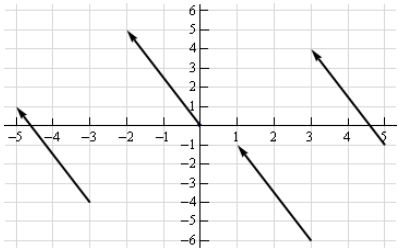


Vector Notation

- A vector can be written more generally as:

$$\vec{v} = \langle a_1, a_2 \rangle$$

  where a_1 and a_2 describe shifts along the axes.
- Now, imagine this vector represents movement from point A to B. In this case, we write the vector as $\vec{AB}$.
- In 2D space, if point A has coordinates $A = (x, y)$ and point B has coordinates $B = (x + a_1, y + a_2)$, the vector $\vec{AB}$ describes the displacement.
- Note the difference in notation between vectors and coordinates.



Vector Notation

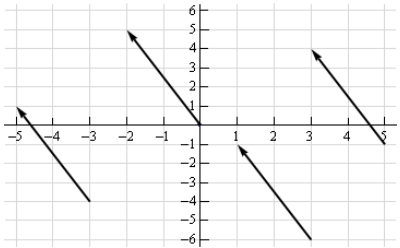
- The direction indicated by the arrow over the vector, $\vec{AB}$, specifies movement from point A to B.
- If the coordinates of A and B are given as $A = (a_1, a_2)$ and $B = (b_1, b_2)$, the vector $\vec{v}$ describing $\vec{AB}$ is:

$$\vec{v} = \langle b_1 - a_1, b_2 - a_2 \rangle$$

- If our vector now starts at B and moves to A, thus $\vec{BA}$, then we can write:

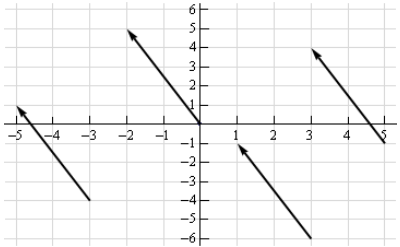
$$\vec{v} = \langle a_1 - b_1, a_2 - b_2 \rangle$$

- Using a **start minus end** approach.



Unit Vectors

- We consider vectors in Cartesian form, that is broken down into components along x and y of a Cartesian domain.
- The component along x is component $\hat{i}$ and along y is component $\hat{j}$.
- Here, $\hat{i}$ and $\hat{j}$ are all called unit vectors and have length equal to 1.



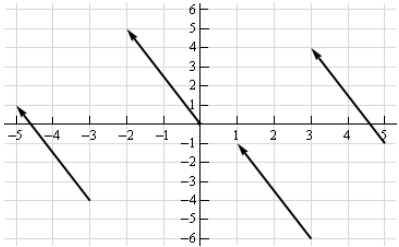
Example: Unit Vectors

- For example, take a look at the following vector:

$$4\hat{i} + 3\hat{j}$$

- For every movement 4 units along the x axis there is 3 units of movement along the y axis.
- Note that this is equivalent to writing

$$\vec{v} = \langle 4, 3 \rangle$$



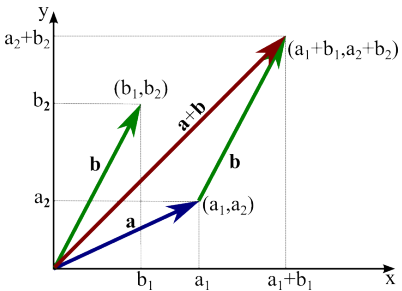
Vector Calculations

- The magnitude of a vector is calculated by taking the square root of all of the square its components (i.e. Pythagoras' theorem):

$$|\bar{a}| = \sqrt{a_1^2 + a_2^2}$$

- Vectors can be added or subtracted (add or subtract by component) or multiplied by a scalar (constant).
- Given the vectors $\vec{a} = \langle a_1, a_2 \rangle$ and $\vec{b} = \langle b_1, b_2 \rangle$:

$$\begin{aligned} \vec{a} + \vec{b} &= \langle a_1 + b_1, a_2 + b_2 \rangle \\ \vec{a} - \vec{b} &= \langle a_1 - b_1, a_2 - b_2 \rangle \\ c\vec{a} &= \langle ca_1, ca_2 \rangle \end{aligned}$$



Example: Vector Calculations

- For example, if $\bar{a} = \langle 4, 3 \rangle = 4\hat{i} + 3\hat{j}$, $\bar{b} = \langle 2, 1 \rangle = 4\hat{i} + 3\hat{j}$ and $c = 2$ then:

$$\begin{aligned} \vec{a} + \vec{b} &= \langle 4 + 2, 3 + 1 \rangle \langle 6, 4 \rangle \\ \vec{a} - \vec{b} &= \langle 4 - 2, 3 - 1 \rangle = \langle 2, 2 \rangle \\ c\vec{a} &= \langle 2 \times 4, 2 \times 3 \rangle = \langle 8, 6 \rangle \\ |\bar{a}| &= \sqrt{4^2 + 3^2} = \sqrt{25} = 5 \end{aligned}$$

Partial Differentiation

- We now think about differentiation of functions of two (or more) variables (i.e. scalar field), $z = f(x, y)$, which now depend on x and y.
- For a one dimensional curve the gradient $\frac{dy}{dx}$ is fairly straight forward.
- When using more than 1D, we have to use something else to denote the gradient.
- The standard that is used is called 'grad', which is defined for a scalar field and can be written ∇f.
- Note that ∇f (where $f = f(x, y, z)$ is a scalar field) is actually a vector.

$$\nabla f \equiv \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} \equiv \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle$$

Partial Differentiation

- The gradient has magnitude and direction!
- Notice the change in the derivative symbol:

$$\frac{\partial f}{\partial x}$$

- Up to this point in our course, we have been using the expression:

$$\frac{df}{dx}$$

- The change is important as this is called a **partial derivative**.
- When you have a function expressed using more than one dimension, this means that when calculating a derivative in say x, you assume other variables such as y or z remain constant.

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- The change is important as this is called a **partial derivative** and we move into using **partial differential equations**.
- When you have a function expressed using more than one dimension, this means that when calculating a derivative in say x, you assume other variables such as y or z remain constant.

Example: Partial Differentiation

- Let's find the partial derivatives of the following function:

$$f(x, y) = 2xy + y^2$$

- If we take y as constant then we calculate partial derivative of f with respect to x:

$$\frac{\partial f}{\partial x} = 2y$$

- The second term disappears here as it has no x within it.
- If we take x as constant then we calculate partial derivative of f with respect to y:

$$\frac{\partial f}{\partial y} = 2x + 2y$$

Example: Partial Differentiation

- Putting this into vector form gives us:

$$\nabla f = \langle 2y, 2x + 2y \rangle$$

- We can use this to calculate the magnitude of the gradient at any given point.

$$\begin{aligned} |\nabla f| &= \sqrt{(2y)^2 + (2x + 2y)^2} \\ &= \sqrt{4y^2 + 4xy + 4xy + 4x^2 + 4y^2} \\ &= \sqrt{4x^2 + 8xy + 8y^2} \end{aligned}$$

Partial Differential Equations

- We could go even further and take second partial derivatives of f to obtain get $\nabla^2 f$.
- This would mean partial differentiating with respect to x and y:

$$\frac{\partial^2 f}{\partial x \partial y}$$

- As we are differentiating twice with respect to all combinations of the variables, $\nabla^2 f$ is a matrix which has amazing properties, but this is outside of this course!
- Integrating partial differential equations analytically can only be done for simple cases and requires knowledge of Fourier methods, which is also outside of this course.
- Nearly all modern weather forecast models use approximate methods to find solutions of **partial differential equations**.